Friedman Test

2026-04-17

Introduction

The Friedman test, introduced by Milton Friedman in 1937, compares three or more paired or repeated measurements without requiring distributional assumptions about the underlying outcomes. It is the rank-based non-parametric analogue of the one-way repeated-measures ANOVA, making it the appropriate choice when the outcome is ordinal, when residuals are markedly non-Normal, or when sample size is too small for the parametric test's distributional assumptions to be defensible. Friedman's test is widely used in sensory-evaluation panels, in within-subject behavioural and clinical studies with small samples, and in agricultural block-design experiments where variance heterogeneity across blocks is a concern.

Prerequisites

A working understanding of rank-based inference, the one-way repeated-measures ANOVA framework, and the relationship between Friedman's test and the Wilcoxon signed-rank test as its two-condition special case.

Theory

For n subjects each measured on k conditions, the observations are ranked within each subject (each subject provides a ranking from 1 to k across conditions). Let R_j be the sum of ranks for condition j across subjects. The Friedman statistic is

$$Q = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1),$$

which is asymptotically χ^2_{k-1} under the null hypothesis of no condition effect. Kendall's coefficient of concordance $W = Q/[n(k-1)]$ in $[0,1]$ provides an effect-size measure; values near 1 indicate strong agreement among subjects' rank orderings of conditions, and values near 0 indicate no consistent ordering. Pairwise post-hoc comparisons typically use the Wilcoxon signed-rank test with Bonferroni correction.

Assumptions

There is one group of subjects each measured on k conditions, the observations are at least on an ordinal scale, and the conditions are pre-specified. The test imposes no Normality or sphericity assumptions, which is its main appeal over parametric repeated-measures ANOVA.

R Implementation

```r
library(rstatix)
set.seed(2026)

n <- 20
time_levels <- c("t0", "t1", "t2", "t3")
df <- expand.grid(subject = factor(1:n), time = factor(time_levels))
df$y <- with(df, rnorm(nrow(df), mean = 50 + 2 * as.numeric(time), sd = 5) +
```

```r
              rep(rnorm(n, 0, 3), times = length(time_levels)))

df |> friedman_test(y ~ time | subject)

df |> friedman_effsize(y ~ time | subject)

df |> wilcox_test(y ~ time, paired = TRUE, p.adjust.method = "bonferroni")
```

Output & Results

The Friedman test on 20 subjects across four time points yields $Q(3) = 46.74$, $p < 0.001$, indicating strong evidence of a time effect. Kendall's $W = 0.78$ classes this as a large concordance — subjects largely agree on the rank ordering of time points. Bonferroni-adjusted pairwise Wilcoxon signed-rank tests then localise the pattern across consecutive time pairs.

Interpretation

A reporting sentence: "The Friedman test showed a significant effect of time on the outcome ($Q_3 = 46.74$, $p < 0.001$), with Kendall's $W = 0.78$ indicating strong concordance among subjects' rank orderings — a large effect. Bonferroni-adjusted pairwise Wilcoxon signed-rank comparisons indicated significant increases between consecutive time points (all adjusted $p < 0.05$). The data are reported on the original scale; the non-parametric test was chosen because residuals from a tentative repeated-measures ANOVA failed the Normality assumption ($p < 0.001$)." Always report effect size.

Practical Tips

- Friedman's test requires complete data across subjects: any subject missing one or more conditions is excluded by listwise deletion. For designs with non-trivial missingness, a linear mixed model with a rank-based or robust variant (`WRS2::rmmcp`) is generally preferable.
- Kendall's $W \in [0, 1]$ measures agreement of rank orderings across subjects rather than effect magnitude on the original scale; an W near 0 means subjects disagreed about which condition was best, while an W near 1 means they agreed strongly.
- For exactly two paired conditions, the Friedman test reduces to (and is equivalent to) the Wilcoxon signed-rank test; the latter is more commonly used in that special case.
- Aligned-rank transform ANOVA (`ARTool::art()`) extends rank-based inference to factorial within-subjects designs, supporting interactions and multiple within-subjects factors that Friedman's test cannot handle on its own.
- For very small samples ($n < 10$), exact permutation-based p-values are preferable to the chi-squared asymptotic approximation; `rstatix::friedman_test()` handles this when an `exact` argument is supplied.
- Friedman's test is sensitive to differences in the location of conditions but assumes the same shape of distribution at each condition; substantial differences in shape can be flagged by the Anderson–Darling or Kolmogorov–Smirnov tests applied across conditions.

R Packages Used

`rstatix::friedman_test()` and `friedman_effsize()` for canonical Friedman analysis with effect size; base R `friedman.test()` for the lightweight implementation; `coin::friedman_test()` for permutation-based exact p-values; `ARTool::art()` for aligned-rank transform ANOVA extending Friedman to factorial designs; `WRS2::rmmcp()` for robust repeated-measures comparisons under non-Normality.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests